

Roll No.

TCS-552

B. TECH. (CSE) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

CLOUD BASED APPLICATION DEVELOPMENT AND MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any **one** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) DTDC Express implemented SAP on AWS with a data warehouse to empower its employees with real-time, self-service reporting tools and achieve a single source of truth for revenue data. DTDC Express Limited is one of the leading express logistics companies in India offering domestic and international services. Evaluate the definition, essential characteristics of Cloud Computing as per NIST along with benefits in the above mentioned organization aspects. 10 Marks (CO1)

OR

- (b) Lion uses AWS to develop and deliver an app and portal rapidly, improve its customer ordering experience and scale on demand to meet fluctuating platform usage. Lion is a leading distributor of beer, wine, spirits and cider in Australia and New Zealand. Evaluate Cluster and Grid Computing and its advantages in the above-mentioned organization aspects. 10 Marks (CO2)

P. T. O.

(2)

2. (a) National Australia Bank is optimizing costs using Amazon EC2 Spot Instances and AWS Graviton processor, contributing to saving an estimated \$1 million per month and engraining cost savings into its company culture. Evaluate Data Intensive Computing and its advantages in the above-mentioned organization aspects.

10 Marks (CO1)

OR

- (b) Active International used the cloud service to enhance its financial planning processes, develop a strategic prospecting application, and launch XACTV, a service that optimizes media buying. These solutions are helping the company expand into new markets. Evaluate Data Intensive Computing and its advantages in the above mentioned organization aspects.

10 Marks (CO2)

3. (a) With Amazon Connect, AAMC has reduced costs by 54% compared to its previous phone system, enhanced its auditing and recording capabilities and gained flexibility and scalability. AAMC provides accident management and assessment services to the Australian insurance industry. Evaluate Compute Services of AWS and its advantages in the above mentioned organization aspects.

10 Marks (CO3)

OR

- (b) With GCP, Centrix achieves nearly 100 percent uptime, minimizes upfront costs and delivers credit response queries in 4 seconds. Centrix is a credit reporting bureau based in New Zealand. Evaluate Application Life Cycle of Google App Engine and its architecture in the above mentioned organization aspects.

10 Marks (CO2)

4. (a) Deputy uses Azure to drive 70 percent faster data request times for customers, scale to support hundreds of millions of data points, save time by eliminating management and maintenance and lower costs. Deputy, based in Australia, provides software that automates scheduling and facilitates workforce management for global customers. Evaluate Azure Compute and Storage services in the above mentioned organization aspects.

10 Marks (CO3)

OR

- (b) Circle of Life migrated to IBM to achieve 25 percent faster deployments and 99.999 percent uptime for its mission-critical ZEVAC platform. Circle of Life's ZEVAC platform analyzes big data on antibiotic prescriptions in hospitals to support data-driven decision-making. Evaluate IBM Cloud services in the above mentioned organization aspects.

10 Marks (CO1)

5. (a) Glossika built its language learning platform on AWS to ensure low latency for users in 148 countries and access to on-demand compute and storage as it expands. Glossika is an educational technology company that uses syntax-sorted and customized content to train people to understand and speak foreign languages. The startup uses Amazon RDS to store lesson data, Amazon ElastiCache for Redis for caching, and Amazon CloudFront as a content delivery network. Evaluate Cloud Authentication Protocols in the above mentioned organization aspects.

10 Marks (CO2)

P. T. O.

OR

(b) Bajaj Energy migrated to SAP on AWS to provision testing instances faster, improve performance by 10-20 percent, and easily create an affordable DR site. Bajaj Energy Limited is a leading thermal power generator in India and a 100 percent subsidiary of Bajaj Group. Evaluate Cloud Federation Stack in the above mentioned organization aspects.

10 Marks (CO1)